Timothy William O'Neil

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Education

Ph.D., Computer Science and Engineering, The University of Notre Dame, May 2002.
- Thesis: *Techniques for Optimizing Loop Scheduling*.
- Advisors: Edwin H.-M. Sha, Peter M. Kogge.

M.S., Computer and Information Sciences, The Ohio State University, June 1993.
M.S., Mathematics, The Ohio State University, August 1991.
B.S., Computer Science, Clarion University of Pennsylvania[1], May 1989.
B.S.Ed., Secondary Education Mathematics, Clarion University of Pennsylvania[1], May 1989.
- Graduated *cum laude*.
- Pennsylvania secondary school (grades 7-12) math certification, June 1989.

Additional Education:
Completed all Ph.D.-related coursework in Mathematics, Bowling Green (Ohio) State University, 1993-96.

Employment Profile, Instructional

2002 – present — Professor (Tenured), Department of Computer Science.
The University of Akron, Akron OH.
- Assistant Professor, 2002-07; Tenured Associate Professor, 2007-13.
- Principle instructor for 1-3 classes per semester.

2001 – 2002 — Graduate Fellow/Research Associate, Dept. of Comp. Sci. and Engineering.
The University of Notre Dame, Notre Dame IN.
- Principle instructor for CSE 210, Spring Semester 2002.
- Assisted Prof. M. Scheutz with curriculum development, Summer 2002.

2000 – 2001 — Senior Lecturer, Department of Computer Science.
The University of Texas at Dallas, Richardson TX.
- Principle instructor for two classes per semester.

1998 – 2000 — Graduate Fellow, Department of Computer Science and Engineering.
The University of Notre Dame, Notre Dame IN.
- Research assistant to Prof. Edwin Sha, 1998-2000.
- Teaching assistant for CSE 341, Spring Semester 2000.

1996 – 1998 — Instructor, Department of Mathematics and Computer Science.
Manchester College[2], North Manchester IN.
- Taught 3-5 classes per semester plus intensive January session.

[1] Now Pennsylvania Western University, Clarion campus.
[2] Now Manchester University.

1993 – 1996 GRADUATE FELLOW, Department of Mathematics and Statistics.
BOWLING GREEN STATE UNIVERSITY, Bowling Green OH.

- Principle instructor for 1-2 classes per semester.

1989 – 1993 GRADUATE TEACHING ASSISTANT, Department of Mathematics.
THE OHIO STATE UNIVERSITY, Columbus OH.

- Principle instructor for 1-4 classes per quarter.

1988 STUDENT TEACHER, Department of Mathematics.
CLARION AREA JUNIOR-SENIOR HIGH SCHOOL, Clarion PA.

- Taught 7th & 8th Grade Math, Algebra and Trig., Fall Semester 1988.

EMPLOYMENT PROFILE, ADMINISTRATIVE

2020 – present CHAIR, Department of Computer Science.
THE UNIVERSITY OF AKRON, Akron OH.

- Founding chair of unit formed by merging Computer Science and Business Information Technology departments and moving new unit to College of Engineering and Polymer Science.
- Deal with all facets of managing the departments, including
    - Personnel matters, including recruitment, retention, promotion, tenure and facilitating faculty development;
    - Fiscal considerations, including salary recommendations, operating budgets and growing endowments;
    - Undergraduate and graduate program, curriculum and course development;
    - Scheduling of course offerings for each academic term;
    - Relations with other University components and external groups; and
    - Student and faculty matters.

2018 – 2020 CHAIR, Departments of Computer Science and Statistics.
THE UNIVERSITY OF AKRON, Akron OH.

- Same as above; duties redefined with University reorganization.

2015 – 2017 COORDINATOR, Department of Computer Science.
THE UNIVERSITY OF AKRON, Akron OH.

- Work directly with the Associate Dean, Natural Sciences Division, College of Arts and Sciences, to maintain day-to-day operation of the Computer Science department.
- Advise Assoc. Dean (who was also serving as interim department chair) on academic and procedural issues related to the department and field.
- Develop schedule of course offerings for each academic term.
- Select, hire, orient, manage, train and evaluate graduate teaching assistants and part-time instructors.
- Assign and supervise grad assistant grading and lab monitoring duties.
- Assist in management of department expenses within budget limitations.

- Provide oversight and answer questions relative to the evaluation of transcripts of transfer students for purposes of admission to the Computer Science programs of study.
- Monitor faculty follow-up with students concerning academic issues.

2012 – 2015 COMPUTER SCIENCE I/II COORDINATOR, Department of Computer Science.
THE UNIVERSITY OF AKRON, Akron OH.

- Designed general course syllabus, as well as more detailed course and lesson plans.
- Managed online delivery of labs using Subversion, Trac and MyProgrammingLab.
- Communicated regularly with other faculty and graduate assistants teaching assigned courses.

TEACHING PROFILE

COLLEGE COURSES TAUGHT, COMPUTER SCIENCE

THE UNIVERSITY OF AKRON (2002 – PRESENT)

- Since assuming chair position, Spring 2018[3]
  - CPSC 316 – Data Structures Sp23, Fa23
    - Cross-listed as CPSC 501 – Fundamentals of Data Structures.
  - CPSC 4/526 – Operating Systems
    Sp18, Sp19, Fa19, Sp20, Fa20, Sp21, Fa21, Sp22, Fa22, Sp23, Fa23, Sp24, Fa24, Sp25, Sp26
  - CPSC 4/577 – Introduction to Parallel Processing Fa18
  - CPSC 490 – Senior Seminar in Computer Science Sp18
  - CPSC 505 – Computing Concepts for Educators[4] Su24, Fa25
  - CPSC 677 – Parallel Processing Sp21, Fa22, Sp24, Fa25
- Prior to assuming chair position, Fall 2002 – Fall 2017
  - 3460:209 – Computer Science I (C++ programming)
    Fa12 (2 secs.), Su13, Fa13 (2 secs), Sp14, Fa14 (2 secs)
    - Previous title: Introduction to Computer Science (Java programming)
      Sp06, Sp07
  - 3460:210 – Computer Science II (C++ programming)
    - Previous title: Data Structures and Algorithms I (Java programming) Fa07
  - 3460:306 – Assembly Language Programming Fa02
  - 3460:316 – Data Structures Fa17
    - Previous title: Data Structures and Algorithms II Sp06
  - 3460:389 – Intermediate Topics in Computer Science: Cloud Computing Fa17
  - 3460:4/521 – Object-Oriented Programming Fa15 (2 secs)
  - 3460:4/526 – Operating Systems
    Fa02, Sp03, Fa03, Sp04, Fa04, Sp05, Fa06, Fa07, Fa08, Fa09, Fa10, Sp13, Fa14, Sp15, Sp16, Sp17
  - 3460:4/540 – Compiler Design Sp08
  - 3460:4/565 – Computer Architecture Sp10, Sp11

[3] Courses relabeled from 3460 to CPSC, Fall Semester 2022.
[4] Taught as CPSC:589 – Topics in Computer Science for first two initial offerings.

- 3460:4/577 – Intro to Parallel Processing Sp03, Fa05, Sp07, Sp09, Fa13, Fa16
- 3460:490 – Senior Seminar in Computer Science Sp04, Sp05, Sp08, Sp09, Sp10, Sp17
- 3460:626 – Advanced Operating Systems Fa09
- 3460:635 – Advanced Algorithms Sp16
  - Previous title: Advanced Algorithms and Complexity Theory Sp11
- 3460:641 – Optimization for Parallel Compilers Fa04, Fa06, Fa08, Sp13, Sp15, Fa17
- 3460:665 – Advanced Computer Architecture Fa05
- 3460:677 – Parallel Processing Fa03, Fa12, Sp14, Fa15, Fa16
  - Assisted with curriculum development, Fall 2011

THE UNIVERSITY OF NOTRE DAME (1998 – 2000, 2001 – 2002)

- CSE 210 – Discrete Mathematics (1 semester)
- CSE 341 – Operating Systems Principles (1 semester, teaching assistant)

THE UNIVERSITY OF TEXAS AT DALLAS (2000 – 2001)

- CS 3305 – Discrete Mathematics for Computing II (3 semesters)
- CS 5333 – Discrete Structures (1 semester)

MANCHESTER COLLEGE (1996 – 1998)

- CPTR 101 – Introduction to Computers (2 semesters)
- CPTR 105 – Computer Programming I (2 semesters, Pascal programming)
- CPTR 217 – C Programming (1 semester)

College Courses Taught, Mathematics

MANCHESTER COLLEGE (1996 – 1998)

- MATH 103 – Survey of Mathematical Thought (1 semester)
- MATH 112 – College Algebra (3 semesters)
- MATH 120 – Precalculus (2 semesters)
- MATH 121 – Calculus I (1 semester)
- MATH 122 – Calculus II (1 semester)
- MATH 130 – Discrete Mathematics (1 semester)
- MATH 233 – Numerical Analysis (1 semester)

BOWLING GREEN STATE UNIVERSITY (1993 – 1996)

- MATH 120 – College Algebra (1 semester)
- MATH 128 – Precalculus Mathematics I (1 semester, team taught)
- MATH 241 – Math for Elementary Teachers I (4 semesters)
- MATH 242 – Math for Elementary Teachers II (1 semester)

THE OHIO STATE UNIVERSITY (1989 – 1993)

- MATH 050 – Precollege Mathematics I (4 quarters)
- MATH 075 – Precollege Mathematics II (5 quarters)
- MATH 104 – Basic College Mathematics (2 quarters)
- MATH 116 – Survey of Finite Mathematics (1 quarter)
- MATH 130 – Mathematical Analysis for Business I (1 quarter)
- MATH 148 – College Algebra (1 quarter)
- MATH 151 – Calculus and Analytic Geometry I (1 quarter, recitation instructor)
- MATH 152 – Calculus and Analytic Geometry II (1 quarter, recitation instructor)

- MATH 153 – Calculus and Analytic Geometry III (1 quarter, recitation instructor)

Research Profile

Primary Research Interests

High level compiler optimization/loop transformation and scheduling, high performance (parallel/distributed) computing, computer science education.

Published Journal Papers

- J.S. Haddad, T. W. O'Neil, A. Deeter, and Z.-H. Duan, "An Implementation of Parallel Bayesian Network Learning", *Journal of Computational Science Education*, Vol. 8, No. 2, July 2017, pp. 24 – 28. DOI: 10.22369/issn.2153-4136/8/2/4
- T.W. O'Neil, "Extended Retiming: Transforming Synchronous Data-Flow Graphs to Minimize Clock Period", *Asian Journal of Applied Science and Engineering*, Vol. 4, December 2015, pp. 201 – 218.
- T.W. O'Neil, "Static Scheduling of Synchronous Data-Flow Graphs Under Resource Constraints", *Parallel and Distributed Computing and Networks*, Vol. 2, 2012. DOI: 10.2316/Journal.211.2012.1.211-1057.
- T.W. O'Neil, D.H. Mugler, K.J. Liszka, R. Gummadi and A.B. Mirza, "Experiments in Parallelizing the Type IV Discrete Cosine Transform", *Computer Technology and Application*, Vol. 3, No. 4, April 2012, pp. 305 – 314.
- T.W. O'Neil, S.F. Khasawneh, M.E. Richter and R.K. Pullaguntla, "Transforming Synchronous Data-Flow Graphs to Reduce Execution Time", *International Journal of Computers and Their Applications*, Vol. 18, June 2011, pp. 111-122.
- E.R. Wheland, T.W. O'Neil, J.D. Adler and K.J. Liszka, "The Poster Session: A Tool for Education, Assessment and Recruitment", *Mathematics and Computer Education*, Vol. 43, No. 2, Spring 2009, pp. 141-150.
- T.W. O'Neil and E. H.-M. Sha, "Time-Constrained Loop Scheduling with Minimal Resources", *Journal of Embedded Computing*, Vol. 2, October 2006, pp. 103-117.
- T.W. O'Neil and E. H.-M. Sha, "Combining Extended Retiming and Unfolding for Rate-Optimal Graph Transformation", *Journal of VLSI Signal Processing Systems for Signal, Image and Video Technology*, Vol. 39, March 2005, pp. 273-293.
- E. H.-M. Sha, T.W. O'Neil and N.L. Passos, "Efficient Polynomial-Time Nested Loop Fusion with Full Parallelism", *International Journal of Computers and Their Applications*, Vol. 10, March 2003, pp. 9-24.
- Z. Wang, T.W. O'Neil and E. H.-M. Sha, "Optimal Loop Scheduling for Hiding Memory Latency Based On Two Level Partitioning and Prefetching", *IEEE Transactions on Signal Processing*, Vol. 49, November 2001, pp. 2853-2864.
- T.W. O'Neil and E. H.-M. Sha, "Retiming Synchronous Data-Flow Graphs to Minimize Execution Time", *IEEE Transactions on Signal Processing*, Vol. 49, October 2001, pp. 2397-2407.
- Z. Wang, T.W. O'Neil and E. H.-M. Sha, "Minimizing Average Schedule Length Under Memory Constraints by Optimal Partitioning and Prefetching", *Journal of VLSI Signal Processing Systems for Signal, Image and Video Technology*, Vol. 27, March 2001, pp. 215-233.

- S. Tongsima, T.W. O'Neil, C. Chantrapornchai and E. H.-M. Sha, "Properties and Algorithms for Unfolding of Probabilistic Data-flow Graphs", *Journal of VLSI Signal Processing Systems for Signal, Image and Video Technology*, Vol. 25, No. 3, July 2000, pp. 215-234.
- F. Chen, T.W. O'Neil and E. H.-M. Sha, "Optimizing Overall Loop Schedules using Prefetching and Partitioning", *IEEE Transactions on Parallel and Distributed Systems*, Vol. 11, No. 6, June 2000, pp. 604-614.
- T.W. O'Neil, "Catalan's Equation", *Mathematics in College*, 1998, pp. 53-58.
- T.W. O'Neil and D. Bhattacharya, "Exploring Fermat's Last Theorem", *Ontario Mathematics Gazette*, Vol. 29, September 1990, pp. 18-21.
- J. Blass, A.M.W. Glass and T.W. O'Neil, "Catalan's conjecture and linear forms in logarithms", *Ulam Quarterly Journal*, accepted but never appeared in print; cited by:
    - C.D. Bennett, L.K. Elderbrock and A.M.W. Glass, "Zero Estimates for Polynomials in Three and Four Variables using Orbits and Stabilisers", in *Hilbert's Tenth Problem: Relations with Arithmetic and Algebraic Geometry*, Amer. Math. Soc., 2000, 357 – 367.
    - M. Mignotte, "Catalan's equation just before 2000", *Number Theory (Turku, 1999)*, de Gruyter, Berlin, 2001, 247 – 254.
    - P. Mihailescu, "A Class Number Free Criterion for Catalan's Conjecture", *J. Number Th.* **99** (2003), 225 – 231.
    - Y.F. Bilu, "Catalan's conjecture (after Mihailescu)", *Seminaire Bourbaki*, 909, 2002-2003; *Astrisque* **294** (2004), 1 – 26.

Published Papers, Refereed Conferences

- T.W. O'Neil, "A Project-Driven Operating Systems Class: A Case Study", in *Proc. 13th International Conference on Frontiers in Education: Computer Science and Computer Engineering*, in conjunction with the *2017 World Congress in Computer Science, Computer Engineering and Applied Computing*, Las Vegas NV, July 2017, pp. 142 – 147.
- T.W. O'Neil and Y. Xiao, "Teaching Parallel Programming Using CUDA: A Case Study", in *Proc. 13th International Conference on Frontiers in Education: Computer Science and Computer Engineering*, in conjunction with the *2017 World Congress in Computer Science, Computer Engineering and Applied Computing*, Las Vegas NV, July 2017, pp. 110 – 115.
- B. Cai, Y. Xiao, T.W. O'Neil and Z.-H. Duan, "Scattered Data Modeling Using GPU: A Case Study", in *Proc. 13th International Conference on Modeling, Simulation and Visualization Methods*, in conjunction with the *2016 World Congress in Computer Science, Computer Engineering and Applied Computing*, Las Vegas NV, July 2016, pp. 173 - 179.
- J.S. Haddad, A. Deeter, Z.-H. Duan and T.W. O'Neil, "Analysis of Parallel Bayesian Networks", in *Proc. ISCA International Conference on Computers and Their Applications*, Las Vegas NV, April 2016, pp. 101 - 106.
- T.W. O'Neil, "Experiments with Massively Parallel Matrix Multiplication", in *Proc. ISCA 29th International Conference on Computers and Their Applications*, Las Vegas NV, March 2014, pp. 135 – 140.
- T.W. O'Neil, "Rate-Optimal Unfolding of Balanced Synchronous Data-Flow Graphs", in *Proc. ISCA 29th International Conference on Computers and Their Applications*, Las Vegas NV, March 2014, pp. 261 – 266.
- T.W. O'Neil, "Unfolding Synchronous Data-Flow Graphs", in *Proc. IASTED 23rd International Conference on Parallel and Distributed Computing and Systems*, Dallas TX, December 2011, pp. 278 – 283.

- T.W. O'Neil, "Resource-Constrained Static Scheduling of Synchronous Data-Flow Graphs", in *Proc. IASTED 23rd International Conference on Parallel and Distributed Computing and Systems*, Dallas TX, December 2011, pp. 124 – 129.
- T.W. O'Neil, A.B. Mirza and D.H. Mugler, "An Experiment in Parallelizing the Fast Fourier Transform", in *Proc. 2011 International Conference on Parallel and Distributed Processing Techniques and Applications*, in conjunction with the *2011 World Congress in Computer Science, Computer Engineering and Applied Computing*, Las Vegas NV, July 2011, Vol. I, pp. 336-341.
- K.C. Chakilam, S.R. Anapalli and T.W. O'Neil, "Minimizing Inter-Iteration Dependencies in Multi-Dimensional Loops", in *Proc. ISCA 22nd International Conference on Parallel and Distributed Computing and Communication Systems*, Louisville KY, September 2009, pp. 25-30.
- S.R. Anapalli, K.C. Chakilam and T.W. O'Neil, "Static Scheduling for Cyclo-Static Data Flow Graphs", in *Proc. 2009 International Conference on Parallel and Distributed Processing Techniques and Applications*, in conjunction with the *2009 World Congress in Computer Science, Computer Engineering and Applied Computing*, Las Vegas NV, July 2009, Vol. I, pp. 302-306.
- R.K. Pullaguntla, S.F. Khasawneh and T.W. O'Neil, "Rotation Scheduling on Synchronous Data Flow Graphs", in *Proc. 2009 International Conference on Parallel and Distributed Processing Techniques and Applications*, in conjunction with the *2009 World Congress in Computer Science, Computer Engineering and Applied Computing*, Las Vegas NV, July 2009, Vol. I, pp. 263-269.
- M.E. Richter, D.J. Poeschl and T.W. O'Neil, "New Heuristics for Rotation Scheduling", in *Proc. 2008 International Conference on Parallel and Distributed Processing Techniques and Applications*, in conjunction with the *2008 World Congress in Computer Science, Computer Engineering and Applied Computing*, Las Vegas NV, July 2008, pp. 10-16.
- S.F. Khasawneh, M.E. Richter and T.W. O'Neil, "Static Scheduling for Synchronous Data-Flow Graphs", in *Proc. ISCA 22nd International Conference on Computers and Their Applications*, Honolulu HI, March 2007, pp. 38-43.
- T.W. O'Neil and E. H.-M. Sha, "Static Scheduling of Split-Node Data-Flow Graphs", in *Proc. IASTED 17th International Conference on Parallel and Distributed Computing and Systems*, Phoenix AZ, November 2005, pp. 125-130.
- T.W. O'Neil and E. H.-M. Sha, "Using Unfolding to Minimize Inter-Iteration Dependencies", in *Proc. IASTED 16th International Conference on Parallel and Distributed Computing and Systems*, Cambridge MA, November 2004, pp. 342-347. (Nominee for best paper.)
- T.W. O'Neil, "Optimal Graph Transformation Assuming Alternate Scheduling Models", in *Proc. ISCA 16th International Conference on Parallel and Distributed Systems*, Reno NV, August 2003, pp. 55-60.
- T.W. O'Neil and E. H.-M. Sha, "Unfolding a Split-Node Data-Flow Graph", in *Proc. IASTED 14th International Conference on Parallel and Distributed Computing and Systems*, Cambridge MA, November 2002, pp. 717-722.
- T.W. O'Neil and E. H.-M. Sha, "Using Retiming to Minimize Inter-Iteration Dependencies", in *Proc. ISCA 16th International Conference on Parallel and Distributed Systems*, Louisville KY, September 2002, pp. 482-487.
- T.W. O'Neil and E. H.-M. Sha, "Minimizing Resources in a Repeating Schedule for a Split-Node Data-Flow Graph", in *Proc. IEEE/ACM 12th Great Lakes Symposium on VLSI*, New York City NY, April 2002, pp. 136-141.

- T.W. O'Neil and E. H.-M. Sha, "On Retiming Synchronous Data-Flow Graphs", in *Proc. ISCA 14th International Conference on Parallel and Distributed Systems*, Richardson TX, August 2001, pp. 103-108.
- T. W. O'Neil, E. H.-M. Sha and S. Tongsima, "Parallelizing Synchronous Data-Flow Graphs via Retiming", *4th International Conference on Algorithms and Architectures*, Hong Kong, December 2000, pp. 252-263.
- T.W. O'Neil and E. H.-M. Sha, "Optimal Graph Transformation using Extended Retiming with Minimal Unfolding", in *Proc. IASTED 12th International Conference on Parallel and Distributed Computing and Systems*, Las Vegas NV, November 2000, Vol. I, pp. 128-133.
- T.W. O'Neil and E. H.-M. Sha, "Minimizing Inter-Iteration Dependencies for Loop Pipelining", in *Proc. ISCA 13th International Conference on Parallel and Distributed Systems*, Las Vegas NV, August 2000, pp. 412-417.
- T.W. O'Neil and E. H.-M. Sha, "Rate-Optimal Graph Transformation via Extended Retiming and Unfolding", in *Proc. IASTED 11th International Conference on Parallel and Distributed Computing and Systems*, Cambridge MA, November 1999, Sec. 10, pp. 764-769.
- T.W. O'Neil, S. Tongsima and E. H.-M. Sha, "Optimal Scheduling of Data-Flow Graphs Using Extended Retiming", in *Proc. ISCA 12th International Conference on Parallel and Distributed Systems*, Fort Lauderdale FL, August 1999, pp. 292-297.
- T.W. O'Neil, S. Tongsima and E. H.-M. Sha, "Extended Retiming: Optimal Scheduling via a Graph-Theoretical Approach", in *Proc. IEEE International Conference on Acoustics, Speech and Signal Processing*, Phoenix AZ, March 1999, Vol. 4, pp. 2001-2004.
- S. Tongsima, T.W. O'Neil and E. H.-M. Sha, "Unfolding Probabilistic Data-flow Graphs Under Different Timing Models", in *Proc. IEEE International Conference on Acoustics, Speech and Signal Processing*, Phoenix AZ, March 1999, Vol. 4, pp. 1889-1892.

Publications Contributed To

- S.R. Anapalli, *Scheduling on Cyclo Static Data Flow Graphs*, Lambert Academic Publishing, 2011, ISBN 9783838373843.
- C.D. Bennett, J. Blass, A.M.W. Glass, D.B. Meronk and R.P. Steiner, "Linear Forms in the Logarithms of Three Positive Rational Numbers", *J. Th. Num. Bordeaux*, Vol. 9, No. 1, 1997, pp. 97-136.
- "January Calendar", *Mathematics Teacher*, Vol. 84, No. 1, 1991, pp. 36-40.
- "December Calendar", *Mathematics Teacher*, Vol. 83, No. 12, 1990, pp. 725-729.

Graduate Students Advised

- Sukumar R. Anapalli, M.S.C.S. (Thesis) student, August 2009. Project: *Static Scheduling and Rotation Scheduling on Cyclo Static Data Flow Graphs*.
- Krishna C. Chakilam, M.S.C.S. (Thesis) student, August 2009. Project: *Representing and Minimizing Multi-Dimensional Dependencies.*
- Neeraj Mogla (co-advisor with L. Lenhart), M.S.C.S. (Project) student, December 2008. Project: *E-read Ohio Online Instructor Application System using Flex and ColdFusion.*
- Ameen B. Mirza (co-advisor with D. Mugler), M.S.C.S. (Thesis) student, December 2008. Project: *Parallel Computation of the Interleaved Fast Fourier Transform with MPI*.
- Rama K. Pullaguntla, M.S.C.S. (Thesis) student, August 2008. Project: *Rotation Scheduling on Synchronous Data-Flow Graphs.*
- Michael E. Richter, M.S.C.S. (Thesis) student, August 2007. Project: *Variations on Rotation Scheduling*.

- Samer F. Khasawneh, M.S.C.S. (Thesis) student, August 2007. Project: *Static Scheduling of Synchronous Data-Flow Graphs*.

Grant Activity

- **P.I. (with L. Pachnowski and K. Plaster), Ohio Department of Education Teach CS College/University Grant, $200,000, 2024 – 2025.**
    - **Extension awarded through June 30 2027.**
- **P.I. (with K. Liszka), NSF DUE 11-23094 RSP #11C128 (Illinois State University), Sub-award #11C128.15, *Service Oriented Paradigm across Introductory Information Technology Curricula*, $5000, 5/1/2013 – 7/31/2014.**
- Senior/Key Person, NIH STTR grant proposal (D. Mugler, P.I.), *Innovative Parallel Implementation of the FFT for Molecular Dynamics*, 2013 – 2014, $300,000.
- Team Member, Ohio Third Frontier University Technology Validation and Start-Up Fund grant (D. Mugler, P.I.), *Parallel Computation of the Fast Fourier Transform*, $100,000 (unfunded).
- **P.I. (with Y. Xiao), NVIDIA CUDA Teaching Center Program, *CUDA Instruction at the University of Akron*, equipment grant (value ~ $17000), 3/1/2011 – 12/31/2012.**
    - **Follow-up equipment grant (value ~$6300), 11/29/2012.**
    - **Follow-up equipment grant (with D. Mugler, value ~$750), 5/6/2014.**
    - **Renewal and follow-up equipment grant (value ~ $1100), 3/8/2016.**
- P.I., Faculty Research Grant, University of Akron, *An Exploration of Rotation Scheduling*, Summer 2011 (unfunded).
- P.I., NSF CCF 10-572, *Efficient Scheduling and Graph Transformations for DSP Applications to Minimize Execution Time*, $153000, December 2010 (unfunded).
- Investigator, HP Technology for Teaching Grant (T. Margush, P.I.), *Tablet Technology: Supporting Programming and Learning*, 2008 (unfunded).
- Investigator, NSF CCLI 05-36761 (T. Margush, P.I.), *Revitalizing the Assembly Language Programming Component of the Undergraduate Computer Science Curriculum*, 2005 (unfunded).
- P.I., Faculty Research Grant, University of Akron, *Extended Retiming via Alternate Methods*, Summer 2003 (unfunded).
- **OSC Cluster Ohio grant, P.I. (with J. Zhu, A. Baldum, Z.-H. Duan, A. Sokolov and P. Wang), *Computational Study of Ligand-Protein-Nanofiber Interactions for the Development of Biosensors*, 16-node Itanium cluster, January 2003.**

Service Profile

Professional Affiliations

- Institute of Electrical and Electronics Engineers, 1999-present.
- Association for Computing Machinery, 2015-present.
- International Society for Computers and Their Applications, 1999-2003, 2007, 2009, 2011, 2014, 2016.
- Mathematics Association of America, 1993-1996.
- National Council of Teachers of Mathematics, 1987-1990.

- Pennsylvania Council of Teachers of Mathematics, 1987-1989.
- Pennsylvania State Education Association, 1986-1989.

Professional Activities

- IASTED International Conference on Parallel and Distributed Computing and Networks.
    - Program Committee, 13th Int'l Conf., February 15-17 2016, Innsbruck Austria.
    - Program Committee, 12th Int'l Conf., February 17-19 2014, Innsbruck Austria.
    - Program Committee, 11th Int'l Conf., February 11-13 2013, Innsbruck Austria.
    - Program Committee, 9th Int'l Conf., February 15-17 2011, Innsbruck Austria.
    - Program Committee, 8th Int'l Conf., February 16-18 2010, Innsbruck Austria.
    - Program Committee, 7th Int'l Conf., February 16-18 2009, Innsbruck Austria.
    - Program Committee, 6th Int'l Conf., February 12-14 2008, Innsbruck Austria.
    - Program Committee, 5th Int'l Conf., February 13-15 2007, Innsbruck Austria.
- IASTED International Conference on Parallel and Distributed Computing and Systems.
    - Program Committee, 24th Int'l Conf., November 12-14 2012, Las Vegas NV.
    - Session Co-chair (Scheduling and Load Balancing), 23rd Int'l Conf., December 14-16 2011, Dallas TX.
    - Program Committee, 23rd Int'l Conf., December 14-16 2011, Dallas TX.
    - Program Committee, 22nd Int'l Conf., November 8-10 2010, Marina del Rey CA.
    - Program Committee, 21st Int'l Conf., November 2-4 2009, Cambridge MA.
    - Program Committee, 20th Int'l Conf., November 16-18 2008, Orlando FL.
    - Program Committee, 19th Int'l Conf., November 19-21 2007, Cambridge MA.
    - Program Committee, 18th Int'l Conf., November 13-15 2006, Dallas TX.
    - Program Committee, 17th Int'l Conf., November 14-16 2005, Phoenix AZ.
    - Program Committee, 16th Int'l Conf., November 8-11 2004, Cambridge MA.
    - Program Committee, 15th Int'l Conf., November 3-5 2003, Marina del Rey CA.
    - Session Co-chair (Compilation Techniques), 14th Int'l Conf., November 4-6 2002, Cambridge MA.
- IFIP International Conference on Embedded and Ubiquitous Computing.
    - Program Committee, 3rd International Workshop on Embedded Software Optimization, in conjunction with EUC 2008, December 17-20 2008, Shanghai PRC.
    - Program Committee, 2006 Int'l Conf., August 1-4 2006, Seoul ROK.
    - Paper Reviewer, 1st International Workshop on Trustworthiness, Reliability and Services in Ubiquitous and Sensor Networks, in conjunction with EUC 2006, August 1-4 2006, Seoul ROK.
- ISCA International Conference on Computers and Their Applications.
    - Session Chair (Medical Applications), 29th Int'l Conf., Mar. 24-26 2014, Las Vegas NV.
    - Session Chair (Algorithms II), 22nd Int'l Conf., March 28-30 2007, Honolulu HI.
    - Program Committee, 19th Int'l Conf., March 18-20 2004, Seattle WA.
- ISCA International Conference on Parallel and Distributed Computing Systems.
    - Session Chair (Algorithms I), 16th Int'l Conf., August 13-15 2003, Reno NV.
    - Program Committee, Session Chair (Mobile Networks and Routing) and Assistant to the Program Chair, 14th Int'l Conf., August 8-10 2001, Richardson TX.
    - Assistant to the Program Committee Co-Chair, 13th Int'l Conf., August 8-10 2000, Las Vegas NV.
- World Congress in Computer Science, Computer Engineering and Applied Computing.
    - Session Chair (Supercomputing, Parallel Processing, Multimedia, VOD, Streaming, Mobile Systems and Algorithms), 2009 International Conference on Parallel and

Distributed Processing Techniques and Applications, in conjunction with WORLDCOMP '09, July 13-16 2009, Las Vegas NV.
    - Session Chair (The WWW and Intranets), 2008 International Conference on Internet Computing, in conjunction with WORLDCOMP '08, July 14-17 2008, Las Vegas NV.
- IARIA International Conference on Systems.
    - Program Committee, 19th Int'l Conf., May 26 – 30 2024, Barcelona Spain.
    - Program Committee, 18th Int'l Conf., April 24 – 28 2023, Venice Italy.
    - Program Committee, 17th Int'l Conf., April 24 – 28 2022, Barcelona Spain.
    - Program Committee, 16th Int'l Conf., April 18 – 22 2021, Porto Portugal.
    - Program Committee, 15th Int'l Conf., February 23 – 27 2020, Lisbon Portugal.
    - Program Committee, 14th Int'l Conf., March 24 – 28 2019, Valencia Spain.
    - Program Committee, 13th Int'l Conf., April 22 – 26 2018, Athens Greece.
    - Program Committee, 12th Int'l Conf., April 24 – 27 2017, Venice Italy.
    - Program Committee, 11th Int'l Conf., February 21 – 25 2016, Lisbon Portugal.
    - Program Committee, 10th Int'l Conf., April 19 – 24 2015, Barcelona Spain.
    - Program Committee, 9th Int'l Conf., February 23 – 27 2014, Nice France.
    - Program Committee, 8th Int'l Conf., January 27 – February 1 2013, Seville Spain.
    - Program Committee, 7th Int'l Conf., Feb. 29 – March 5 2012, St. Gilles Reunion Island.
- Local Organizing Committee Co-chair, 6th International Conference on Rough Sets and Current Trends in Computing, October 23-25 2008, Akron OH.
- Program Committee, IEEE 5th International Symposium on Embedded Computing, October 6-8 2008, Beijing PRC.
- Program Committee, ISCA 17th International Conference on Computer Applications in Industry and Engineering, November 17-19 2004, Orlando FL.
- Program Committee, 2000 International Conference on Information Technology for the New Millennium, Dec. 18-20 2000, Bangkok Thailand.
- Reviewer for many journals including *The Computer Journal*, *International Journal of Computers and Applications*, *IEEE Transactions on Parallel and Distributed Systems*, *Journal of VLSI Signal Processing*, *IEEE Transactions on Signal Processing*, *IEEE Transactions on Circuits and Systems*, *IEEE Transactions on Computers*, *Journal of Supercomputing*, etc.
- Reviewer for discrete math textbook by Thomas Koshy (2002), Java textbook by Elizabeth Boese (2006) and compilers textbook by Fischer and LeBlanc (2008).

University/College Service Activities

University of Akron (2002-present)

- Chair, Dept. Reappointment Committee, Academic Year 2012-2013.
- Dept. Honors Faculty Advisor, Academic Years 2012-2013 through present.
- Dept. Graduate Admissions Committee
    - Chair, Academic Years 2009-2010 and 2010-2011.
    - Member, Academic Year 2012-2013.
- Member, Buchtel College of Arts and Sciences Online Learning Committee/Technology Working Group, Spring Semester 2013.
- Dept. Graduate Faculty Chair, Academic Years 2008-2009, 2009-2010 and 2010-2011.
- Buchtel College Council.
    - Department representative, 2005-06 and 2006-07 Academic Years.
    - Member, Curriculum Committee, 2012-13 Academic Year.
- Coordinator

    - Dept. *Ad Hoc* Committee for Revising Retention/Tenure/Promotion Guidelines, Spring 2010.
    - Dept. *Ad Hoc* Committee for Revising Chair Review Guidelines, Spring 2008.
    - Spring 2007 "C.S. Day" event.
    - Fall 2005 and Fall 2006 "Welcome Back" events.
- Member, Department Faculty Merit Advisory Committee, Summers 2006 and 2007.
- Search Committee Member
    - Chair, Computer Science Department, Spring 2009.
    - Systems Administrator, Computer Science Department, Spring 2008.
    - Assistant Dir. Employer Relations, Center for Career Management, Spring 2005.
    - Assistant Professor, Computer Science Department, Fall 2003, Springs 2011, 2012 and 2013.
- Conference on Undergraduate and Graduate Student Research (CUGSR)
    - Judge, Outstanding Student Research Award, March 2011.
    - Judge, poster session, March 2010.
    - Moderator, CS/EE session, March 2009.
- Commencement exercises.
    - Department Representative, Springs 2003 and 2013.
    - Assistant marshal, Summers 2006, 2009 and 2012.
- Reviewer, Kent State University Master of Science in Digital Sciences proposal, Spring 2011.
- University Representative, Ohio Supercomputer Center (OSC) State Users Group, Academic Year 2002-2003.

MANCHESTER COLLEGE (1996-98)

- Dept. Rep., FESP (First-year Experience in the Sciences Program) Committee, 1996-98.
- Faculty Advisor, Student Senate, 1996-98.
- Member, Dept. *Ad Hoc* Committee for Revising Comp. Sci. Major and Minor, 1996-97.
- Member, Dept. *Ad Hoc* Committee for Precalculus textbook selection, 1997.
- Proctor, Math Placement Exams for incoming first-year students, Summers 1997 and 1998.
- Academic advisor to incoming first-year students, Summer Orientation 1997 and 1998.
- Dept. Rep., Campus Visitation Days, 1997-98. (Recruited prospective students).
- Interviewer, Scholarship Days, 1996-98. (Reviewed student eligibility for scholarships.)
- Coordinator for Campus Visit by DPS, Inc. of Indianapolis, October 1997.